Semester III **B. Tech CSE (AI)**

BCSAI 306: COMPUTATIONAL SYSTEMS BIOLOGY

2L + 0T + 0P + 2C **MM 100**

Unit 1: Introduction to Mathematical Modelling

Collection, Classification and Tabulation of data, Bar diagrams and Pie diagrams, Histogram, Frequency curve and frequency polygon, Ogives. Mean, median, mode, Standard deviation.

Unit 2: Introduction to Static Networks

Graph Theory, Overview of types of Graphs-cyclic/acyclic,complete bipartite, Graph Terminology ,Representation of Networks, Sparse Matrices, Basic Network parameters

Unit 3: Network Biology and Applications

Biological Problems map to graph problems, Networks Jargons-Node, Density, Degree, Clustering Co-efficient, Closeness, Connected Component, Motifs

Unit 4: Reconstruction of Biological Networks

Random Networks, Small-world networks, power law networks, Centrality-lethality Analysis, Assortavity, Community Detection, Protein Network, Gene Regulatory networks

Unit 5: Dynamic Modelling of Biological systems

Introduction, Biochemical Network Model-Michaelis-Menten Kinetics, Non-biological network model-planetary motion,stock-market behavior

Text/Reference Books

1. Advanced Engineering Mathematics, Erwin Kreyszig, John Wiley and Sons, 2001.
2. Computational Cell Biology, Christopher Fall, Springer, 2000.
3. Mathematical models in biophysics, Riznichenko Galina Yur'evna, Book Online, Biophysical society.
4. Voit E (2012) A First Course in Systems Biology. Garland Science, 1/e. ISBN 0815344678
5. Klipp E (2009) Systems biology: a textbook. Wiley-VCH, 1/e. ISBN 9783527318742
6. Newman MEJ (2011) Networks: an introduction. Oxford Univ. Press. ISBN 9780199206650